

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Technology (CSE)

QUIZ #2 (SET-B)

DURATION: 30 MINUTES

SUMMER SEMESTER, 2023-2024

FULL MARKS: 15

CSE 4711: Artificial Intelligence

Programmable calculators are not allowed.

Answer **all 2 (two)** questions. Figures in the right margin indicate full marks of questions whereas corresponding CO and PO are written within parentheses.

Student ID:

1. Pacman wants to try some food around IUT. There are three restaurants $\{A, B, C\}$ and two juice stands $\{D, E\}$ that he wants to visit. But he only has four hours to do so. So he divides his time into four time slots $\{1, 2, 3, 4\}$, each consisting of an hour. He can visit multiple shops within the same time slot. Pacman has to follow the following constraints:

- Two restaurants can not be visited in the same time slot.
- Two juice stands can not be visited in the same time slot.
- Restaurant A does not open for the first time slot.
- Restaurant C must be visited after A and B .
- Restaurant A and juice stand D must be assigned to the same time slot.

- a) Formulate the problem as a CSP by identifying the variables, their domains, and the constraints.

5
(CO1)
(PO1)

Solution:

Variables: $\{A, B, C, D, E\}$, **Domains:** $A \in \{1, 2, 3, 4\}$, $B \in \{1, 2, 3, 4\}$, $C \in \{1, 2, 3, 4\}$, $D \in \{1, 2, 3, 4\}$, $E \in \{1, 2, 3, 4\}$

Constraints:

- $A \neq B, A \neq C, B \neq C$
- $D \neq E$
- $A \neq \{1\}$
- $C > A$ and $C > B$
- $A = D$

Rubric:

- 1 point for correctly stating the variables.
- 1 point for correctly stating the domains.
- 3 point for correctly stating the constraints.

- b) Pacman wants to find a valid arrangement by using backtracking search. Suppose all variables are unassigned initially. Pacman first assigns $\{2\}$ to the variable A and enforces arc consistency. What values remain in the domain of each variable?

4
(CO1)
(PO1)

Solution:

- $A = \{2\}$
- $B = \{1, 3, 4\}$
- $C = \{3, 4\}$
- $D = \{2\}$
- $E = \{1, 3, 4\}$

Rubric:

- 1 point for correctly identifying the domain of each variable, ignoring A as it is already given.

2. Consider a CSP with $L + R$ variables. There are d possible assignments to each variable. The CSP only has binary constraints. If we remove one of the binary constraints, then the CSP can be decomposed into two independent subproblems: A *left* CSP (V_L) with L variables and a *right* CSP (V_R) with R variables. Let, $X \in V_L$ and $Y \in V_R$ be the variables that are connected by the only constraint that joins V_R and V_L . 6
(CO1)
(PO1)

Without removing any constraints, we assign a value to one of the variables in V_L and run forward checking. What is the **maximum** number of variables whose domains could be changed as a result of this?

Solution:

The domains will only change if the variable is connected to the variable that is being changed. A variable on the left side could be connected to at most $L - 1$ variables belonging to V_L in addition to the 1 variable belonging to V_R . So the most number of variables that could be affected is L .

Rubric:

- 3 points for each correct answer.